SWAT4HCLS 2026

Tutorial Proposal

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Title

Making GraphRAG work for the Healthcare and Life Sciences: Ensuring Knowledge Graph Quality for Grounded LLMs

Abstract

Large Language Models (LLMs) are increasingly being combined with knowledge graphs (KGs) through Graph-based Retrieval-Augmented Generation (GraphRAG) to produce grounded, accurate, and explainable outputs. In domains such as healthcare and life sciences, where factual precision and interpretability are critical, the quality of the underlying knowledge graph becomes a decisive factor for success. However, many knowledge graphs used in such pipelines, especially those auto-generated from text or integrated from heterogeneous biomedical datasets, suffer from issues of incomplete structure, ambiguous semantics, or misaligned identifiers. These problems weaken retrieval performance, lead to misleading or hallucinated responses, and make validation difficult.

This half-day, hands-on tutorial introduces a practical framework for evaluating and improving the quality of knowledge graphs used in GraphRAG applications. Through real-world examples of biomedical applications and ontologies/knowledge graphs, participants will learn:

1. How GraphRAG systems work and how to develop and evaluate simple GraphRAG pipelines with RDF knowledge graphs.

2. How knowledge graph quality affects the effectiveness of GraphRAG systems and what common quality issues to be wary of.

3. How to apply practical methods and heuristics for systematically detecting quality issues in knowledge graphs and linked data.

4. How to design strategies to improve graph structure and content for better AI performance.

Audience

Knowledge graph and linked data practitioners, data architects, and modelers, as well as AI/ML professionals seeking to integrate semantic data into generative AI workflows.

Prerequisite knowledge

Basic familiarity with knowledge graph concepts and/or LLM-based applications is recommended. No prior coding or GraphRAG implementation experience required.

Structure and tentative schedule (3 hours total)

Format: Interactive lecture with guided walkthroughs, collaborative analysis, and discussion.

Number of participants: Typically 15-30, though no maximum

Tentative agenda and schedule

Session 1: Welcome and Context Setting (15 min)

- Quick intro to GraphRAG: what it is and why it matters
- Goals and structure of the tutorial

Session 2: Building GraphRAG Systems (45min)

- What are LLMs and what are their strengths and weaknesses
- What is RAG and GraphRAG
- GraphRAG architectures and patterns
- *Hands-on demo: Building a GraphRAG system for question answering over biomedical knowledge*

Session 3: Evaluating GraphRAG Systems (30min)

- RAG Evaluation metrics and how to use them
- *Hands-on: Using DeepEval to evaluate the Graph RAG system*
- Knowledge Graph quality issues and how they affect RAG

Break

Session 4: Assessing Knowledge Graph Quality (45min)

- Key quality dimensions
- Common quality issues and causes
- Quality assessment approaches and techniques
- *Hands-on demo: Automatically detecting quality problems in knowledge graphs*

Session 5L Improving Knowledge Graph Quality (30 min)

- Improving the knowledge graph
- Improving the knowledge graph construction and maintenance process

Session 6: Wrap-up & Takeaways(15min)

- Summary of key insights
- Discussion and Q&A

Tutor

Panos Alexopoulos is a data and AI practitioner, author, and educator, with nearly two decades of experience in knowledge graphs, ontology engineering, and data semantics. As of April 2025, he works as Lead Semantic Data & AI Solutions at Triply, a semantic data integration company, after nine years as Head of Ontology at Textkernel in Amsterdam, where he led the development of a large cross-lingual knowledge graph in the HR and recruitment domain. Panos holds a PhD in Knowledge Engineering and Management from the National Technical University of Athens and he is the author of *Semantic Modeling for Data: Avoiding Pitfalls and Breaking Dilemmas* (O'Reilly, 2020), a practical guide for building high-quality semantic data models. HeHe is also a seasoned educator, having delivered several courses and training programs on the topic of knowledge graphs and large language models.

Contact

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